

More bio-inspired drones...

- ▶ The US Army is working on Robo-Fly, which will one day relieve humans from the life-threatening job of espionage. Using piezoelectric materials, they have developed a fruit-fly-size micro-robot that can flap its wings without a motor. The prototype wings that they have made are just 3mm long, and are made of lead-zirconium-titanate, a material that bends and flaps when voltage is applied to it. They have also used the same material to design a set of tiny robotic legs (like a millipede's) that can crawl when voltage is applied to these. It might be 10 to 15 years before these wings and legs actually turn into spying insects.
- ▶ Mirko Kovac, director of Aerial Robotics Laboratory at Imperial College London, believes in turning to nature for answers to drone challenges. By studying the perching of birds, he has developed a pigeon-size drone that can fly to the side of a building and comfortably perch on it. Basically, mechanical forces resulting from the impact cause the drone's two arms to fly forward and make a grabbing action that holds onto the brickwork without spending any

extra energy. He is now studying the multi-modal mobility of birds and other creatures, which comfortably move between water, land and sky, to develop drones with similar capabilities.

- ▶ David Lentink and his team at Stanford University, USA, have designed bird-inspired wings for drones. Unlike the rigid wings of helicopters and fixed-wing aircraft, these special wings can flap like a bird's and fold back on impact, enabling these to recover instantly after a collision and continue flying like birds do. The wings are made of carbon-fibre and Mylar film. Each wing consists of two parts, one arm wing and one hand wing, hinged together with a 3D-printed wrist joint that enables the hand wing to fold back over the arm wing. The arm wing is further attached to the body by a shoulder joint. When the wings flap, a centrifugal force is created, enabling smooth flight. On collision, the wings fold back without any damage, recover instantly and start flapping again. All this happens passively, without requiring any electronics. This would make drones lighter and more reliable, too.

for UAVs. The artificial eye measures just two cubic millimetres and weighs just two milligrams. It features a lens on top of all three electronic photodetectors in a triangular pattern. The device combines the measurements of these photodetectors to determine the speed and direction of any motion in its field-of-view. This is similar to how the segmented eyes of insects capture their environment to avoid obstacles.

The team has also developed algorithms to process signals from these devices, which will be programmed into onboard chips to compute relevant parameters like distance to objects, time until potential collision and so on.

The artificial eye has been tested under varied conditions like poorly-lit rooms, bright and sunny outdoor spots, and it seems poised to outdo the original itself, as results show that it is able to detect mo-

tion three times faster than actual flying insects.

Autonomy is a matter of perception

If drones are to be really useful, delivering parcels, spying enemy territories, entering danger zones ahead of fire-fighters and first-responders, and more, these need to be aware of their environment and capable of flying autonomously. Drone makers across the world are investing

Other efforts to make drones self-aware...

- ▶ When drones suddenly run out of power or lose the global positioning system (GPS) signal, these crash land. In order to prevent such situations, University of Zurich has developed drones with improved safety features. The drones have a camera, acceleration sensors and an orientation system that emulates the human visual system and sense of balance. When a failure situation is detected, the drone tries to regain balance and, when that is not possible, it surveys its surroundings, builds a 3D model of the environment and tries to find a safe place to land. All image processing and control runs on a smartphone processor onboard the drone, which enables the drone to act independently without requiring instructions from an operator.
- ▶ Last year, DJI launched a new multi-modal sensing system called Guidance, which empowers drones like the company's Matrice 100 with autonomous obstacle-avoidance capabilities. Guidance consists of an array of five ultrasonic rangefinders, a set of integrated visual cameras running advanced algorithms and an onboard processor to make sense of all data. Guidance enables drones to hover in place, maintain their positions and avoid obstacles, without GPS support.
- ▶ Andrew Barry, a PhD student at Massachusetts Institute of Technology (MIT), USA, recently developed an obstacle-

avoidance system for drones, which uses only two mobile phones worth of onboard computing hardware and real-time image processing, according to an Institute of Electrical and Electronics Engineers (IEEE) report. The solution is based on stereo filtering from a pair of 376 x 240 pixel resolution, 120 frames-per-second cameras spaced 34 centimetres apart. The drone focuses its attention on pixels that are about ten metres away and nothing else. It saves these pixels in its memory and the next image adds more pixels to it, gradually helping the drone to build a 3D map of what lies ahead. This technique is called pushbroom stereo detection. With a little bit of ironing out, this method could help drones to fly autonomously, at lower costs.

- ▶ While remote-controlled drones are capable of flying at great speeds, autonomous ones still move quite slowly in order to make the required obstacle-avoidance calculations. However, birds and insects are able to fly very fast, without crashing into anything. This fact has inspired Defence Advanced Research Projects Agency (DARPA) to fund the development of small, lightweight and autonomous drones that can fly at speeds greater than 70kmph, manoeuvring adeptly to avoid obstacles. Rising to the occasion, Draper Labs and a group at MIT have started work on software systems that will help avoid obstacles at such high speeds.